John D. Duncan, III

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Motivation

Making things that are useful to people and that work well.

Employment

Quorum Analytics — Washington, DC

2022–2024 | Senior Software Engineer (50 to 70+ hours/week)
2020–2022 | Software Engineer II
2018–2020 | Software Engineer

Python, Django, PostgreSQL, React, JavaScript, TypeScript, HTML, CSS, D3, Redux, React Native, PostGIS, Elasticsearch

Wrote and launched over a dozen flagship products for Quorum's industry-leading civic tech and public affairs platform, consistently solving the team's most complex technical challenges while delivering projects on time and under budget.

- Instrumental in helping scale ARR from $9M to $64M+.
- Authored 5k+ commits, 1k+ code reviews, 700+ PRs, and conducted 100+ technical interviews.
- Led cross-functional teams in collection, organization, and presentation of legislative, regulatory, compliance, financial, geospatial, and user-generated datasets.

Leadership

- Discovered bug in Google Chrome URL parser
- Wrote design system
- Created full-stack guild
- Led Quorum Accelerate apprenticeship program
- Mentored dozens of engineers
- Onboarding coach
- Rockstar of the Quarter

Projects

PAC Hybrid	Technical Lead	Team of 10
Quorum PAC	Technical Lead	Team of 8
Search UI/UX	Technical Lead	Team of 4
Dashboards	Technical Lead	Team of 2
Transcripts	Front-end Lead	Team of 2
Bulk Actions	Technical Lead	Team of 1
International	Front-end Lead	Team of 2
Regulations	Front-end Lead	Team of 1
Search	Engineer	Team of 2

Duties

- Mentorship, coaching, onboarding
- Interviews (technical, design)
- Tests (regression, integration, unit)
- Code review
- Technical Design Documents
- Product Requirements Documents
- "Learning Lab" presentations
- 24hr on-call shifts
- Daily deployment

Massachusetts Institute of Technology: Lincoln Laboratory (MIT LL) — Lexington, MA

Summer 2016 | Research Intern (40 hours/week)

Java, Android SDK, Google Maps SDK, JavaScript, ExtJS, HTML, CSS, SQL, Acme (Plan 9)

Wrote Local Evacuation Alert Verification (LEAV, sponsored by DHS) program for HURREVAC-eXtended (HVX, sponsored by FEMA) with the Humanitarian Assistance and Disaster Relief Systems (HADR, Group 44) at MIT LL on Hanscom AFB.

- Designed and built LEAV, an Android application that routes evacuees to shelters, visualizes live evacuation zone status through an interactive map, and enables communication between Emergency Managers and evacuees.
- Developed an interactive JavaScript module for HVX that enables Emergency Managers to dynamically update and monitor the status of evacuation zones in a web interface.
- Reduced WFS-T request size by an order of magnitude.

Technical Skills

Languages	Python, JavaScript, TypeScript, Go, SQL, LaTeX	**SCM & CI/CD**	Git, Github, Jenkins, Travis
Databases	PostgreSQL, PostGIS, Elasticsearch, SQLite	**Infrastructure**	Docker, Kubernetes, Ansible
Web	Django, React, Next.js, D3, HTML5, CSS3	**Observability**	Sentry, Datadog, Grafana
Devops	AWS (EC2, RDS, S3, Lambda, DMS, Route53)	**Operating Systems**	macOS, Plan 9, Linux
	Atlassian (Jira, Confluence, Opsgenie)		

Education

Gettysburg College — Gettysburg, PA

Bachelor of Science in Computer Science (3.60, Honors) and Philosophy (3.73, Honors) — 2013–2017

Computer Science	**Capstone** *Adam's Wellness Family Connection*	Advisor Dr. Rod Tosten
Class of 2017	Outstanding Computer Science Student Award	
Philosophy	**Thesis** *Virtual Futures: Virtuality as Political Praxis*	Advisor Dr. Lisa Portmess
	Capstone *Infinite Horizons: The World to Mind Question*	Advisor Dr. Daniel DeNicola
Extracurricular	Eisenhower Institute Undergraduate Fellow (one of eight)	Advisor Dr. Shirley Warshaw
2016–2017	Gettysburg Political Philosophy Collective	Co-President
2016–2017	Gettysburg College Independents	Treasurer
2015–2017	Gettysburg Association for Computing Machinery (ACM) chapter	Secretary, Vice President

Professional Projects

Quorum Analytics Washington, DC

PAC Hybrid | 2024
Position | Technical lead (search, dashboards, back-end, SQL) & team leader
Team | 8 engineers, 2 product managers, 2 QA engineers, 2 engineering managers, 1 engineering director
Technologies | Django, Python, React, JavaScript, PostgreSQL, MySQL, Kubernetes, AWS RDS/DMS

Summary Designed, implemented, and launched the Quorum PAC Hybrid Federal Election Commission (FEC) reporting platform, integrating Cision's market leading Government Relations PAC Product into the Quorum PAC federal filing platform and ensuring client compliance with evolving FEC financial regulatory enforcement requirements and reporting procedures. Wrote the full-stack Search infrastructure and MVP for the full-stack Dashboard integration.

- Designed and implemented a novel search infrastructure enabling disparate filters to share values, meeting critical functional requirements and ensuring product viability.
- Developed optimized O(1) QueryMethods to support complex summation and grouping over arbitrarily filtered PAC Transaction datasets in search and dashboards, replacing an inflexible yearly contribution caching model, enabling efficient searches across four datasets, and improving performance and scalability.
- Discovered and implemented an undocumented Django feature, reducing PAC Hybrid search database query response time from 60 seconds to 20 milliseconds and boosting overall performance by an order of magnitude. Optimization ensured adherence to contractual SLOs, maintaining product viability for clients managing hundreds of thousands to millions of transactions. Additionally, resolved a decade-old performance issue affecting all queries involving partitioned table joins.
- Conducted legal research by auditing 50+ years of FEC advisory opinions, identifying discrepancies and inconsistencies in the FEC's official list of national party committee C-numbers listed across the FEC website, advisory opinions, and the fec-cms GitHub repository. Research ensured client compliance with critical regulatory requirements due to elevated contribution limits for FEC-designated National Party Committees.

Quorum PAC | 2021–2023
Position | Technical Lead (search, dashboards, FEC submission, forms, back-end, SQL)
Team | 8 Engineers
Technologies | Django, Python, React, JavaScript, PostgreSQL, Docker, Kubernetes, AWS RDS

Summary Designed and implemented full-stack search and dashboard infrastructure, an FEC submission microservice using FEC's VenPak software, and forms for state political committees, charity matching, and check processing in the new Quorum PAC FEC reporting platform, enabling Quorum's entry into the $15M+ PAC software market and helping clients remain compliant, raise funds, and report results to the FEC.

Search UI/UX | 2021
Position | Technical Lead
Team | 4 engineers (2 apprentices), 1 designer, 1 engineering manager
Technologies | Django, Python, React, JavaScript, PostgreSQL, AWS RDS

Summary Gathered requirements, designed, and implemented functional and UI/UX enhancements for Quorum's search feature, the default homepage for all users. Mentored two apprentice engineers transitioning from non-engineering careers by providing constant guidance and facilitating their technical growth. Wrote the project technical design document (TDD), collaborated closely with product and design teams, and developed reusable design system components. Delivered key new functionality, including arbitrary filter and filter group searching by name, value, placeholder, or filter options; novel ordering system enabling users to sort 15 datasets by 27 unique database columns; redesigned global search pop-down, allowing seamless access to 37 datasets from any Quorum feature or location.

Dashboards | 2019–2020
Position | Technical Lead
Team | 2 engineers, 1 designer, 1 engineering manager
Technologies | Django, SQL, Python, React, D3, react-grid-layout, Javascript, Postgres, AWS RDS

Summary Developed an industry-first, full-stack Dashboard feature empowering non-technical users to analyze and visualize data through an intuitive interface, enabling visualization, filtering, and grouping of key metrics and performance indicators and centralization of information for tracking progress, identifying trends, and evaluating performance benchmarks.

- Designed and implemented 9 widget types (e.g., List, Calendar, Visualization, etc.) to address diverse user needs.
- Wrote 73 Django SQL query methods spanning 25 datasets for complex column, date, and sum grouping for visualizations.
- Created portable D3-based visualization components (Bar, Pie, Line, and Map) utilized in dashboards and other Quorum tools such as search visualizations and official district maps.
- Engineered a novel GraphQL-like data shaping system for efficient data storage, serialization, and caching.
- Identified and resolved critical bugs in Chromium URI parsing and react-grid-layout to enhance dashboard performance.

Transcripts | 2019
Position | Front-end Lead
Team | 2 engineers, 2 engineering managers, CTO/co-founder
Technologies | Django, Python, React, JavaScript, PostgreSQL, AWS S3/RDS

Summary Developed the front-end for the Committee and Presidential Transcripts feature, enabling users to search, download, and visualize transcripts of congressional hearings and presidential speeches within an hour of completion.

Implemented bi-directional video-transcript synchronization, which enabled auto-scrolling and highlighting of transcript statements during video playback or manual scrubbing and added functionality to sync video playback to specific timestamps when transcript statements are clicked.
Designed and built a committee profile overview page which included rewriting a non-performant API endpoint using a novel QueryMethod to support arbitrary filtering of database columns by enum attributes, significantly improving performance and scalability.

Bulk Actions	2019
Position	Technical Lead
Team	1 engineer, 1 engineering manager, CTO/co-founder
Technologies	Django, SQL, Python, React, JavaScript, PostgreSQL, AWS RDS

Summary Designed and implemented a full-stack Bulk Action feature enabling users to select and update up to 1 million rows across 17 datasets with 13 action types. Researched and optimized batch updates using Django ORM and raw SQL, culminating in O(1) SQL UPDATEs for scalable performance. Delivered 13 robust bulk actions, supporting high-efficiency client data processing and mutation with no reported bugs since launch in 2018. Enhanced user productivity by enabling seamless, large-scale updates in a single operation.

International	2018
Position	Front-end Lead
Team	2 engineers, 1 engineering manager, CTO/co-founder
Technologies	Django, SQL, Python, React, JavaScript, PostgreSQL, AWS RDS

Summary Scaled Quorum product to 2 new continents and dozens of new countries by writing the front-end for the international search feature, which allows clients to gather intelligence, identify trends, and communicate with key international stakeholders. I modified the existing official, legislation, and regulation search datasets to support filtering and visualizing international data.

Regulations	2018
Position	Front-end Lead
Team	6 engineers, 1 engineering manager, CTO/co-founder
Technologies	Django, SQL, Python, React, JavaScript, PostgreSQL, AWS RDS

Summary Wrote the front-end search and profile for the regulations feature, which gives clients access to comprehensive state and federal regulatory data and tracking tools. My work included dozens of filters and QueryMethods, support for alerts, lists, and downloads, and ability to set stance, priorities, and issues. I rewrote the existing Bill timeline to support Regulations and other arbitrary datasets in the future.

Search	2018
Position	Engineer
Team	2 engineers, 1 engineering manager, CTO/co-founder
Technologies	Django, SQL, Python, React, JavaScript, PostgreSQL, AWS RDS

Summary Re-wrote infrastructure and expanded functionality for search feature, which is the default homepage for all Quorum clients. Updated 30+ datasets with new information cards, filters, csv downloads, and visualizations. Contributed toward shared hybrid infrastructure which powered search on both the desktop React application and React Native mobile application. Due to unique functional requirements, I researched the React internals and HTML specification and wrote a library which converts raw HTML into React Elements by recursively processing DOM nodes and converting them into React elements.

Leadership

The Cairo Board of Directors Washington, DC

Co-led a comprehensive board transformation to eliminate deficit in $1M+ annual budget. Implemented a strategic financial management plan by auditing finances, cutting wasteful spending, restructuring staff, strengthening vendor accountability, and developing an aggressive capital budget improvement strategy—all while keeping fee increases under 5%.

Partnered with a horticulturist from Harvard's Dumbarton Oaks Gardens and a landscape architect from Oehme, van Sweden & Associates to design and install heirloom perennial pollinator garden for The Cairo, a historic 1894 building listed on the DC Inventory of Historic Sites (1990) and the National Register of Historic Places (1994).

Personal Projects

In my free time, I enjoy challenging myself by working on idiosyncratic projects across diverse domains. Recently, I built a Golang query builder compatible with Django ORM syntax, uncovered a bug in the Google Chromium URL parser, discovered a novel method to losslessly convert legacy Apple Macintosh data fork fonts to macOS resource forks, ported James Ashton's X-Face decoder from C to Golang, designed a JavaScript weather app using only bitmaps from the 1980s/90s, created a 3x4 pixel font, and wrote a Mozilla XUL/XPCOM add-on to display 'face' images from email and newsgroup message headers. I also have a passion for design and previously won an interface design competition.